

Research Notes on Autoregressive model

Autoregressive model

>> Section 1

In statistics, an autoregressive (AR) model is a modelled representation of a type of random process. It can be used to describe time-varying processes from many natural and artificial sources. The model specifies output variables that are dependent linearly on their own previous values on a stochastic basis. The model is in the form of a stochastic difference equation (or recurrence relation) which should not be confused with a differential equation. Together with the moving-average (MA) model, it is a special case and key component of the more general autoregressive–moving-average (ARMA) and autoregressive integrated moving average (ARIMA) models of time series, which have a more complicated stochastic structure; it is also a special case of the vector autoregressive model (VAR), which consists of a system of more than one interlocking stochastic difference equation in more than one evolving random variable. Another important extension is the time-varying autoregressive (TVAR) model, where the autoregressive coefficients are allowed to change over time to model evolving or non-stationary processes. TVAR models are widely applied in cases where the underlying dynamics of the system are not constant, such as in sensors time series modelling, climate science, economics and finance (as econometrics), signal processing, telecommunications, radar systems, and biological signals. Unlike the moving-average (MA) model, the autoregressive model is not always stationary; non-stationarity can arise either due to the presence of a unit root or due to time-varying model parameters, as in time-varying autoregressive models. Large language models are called autoregressive, but they are not a classical autoregressive model in this sense because they are not linear.

>> Section 2

where $\phi_1, \dots, \phi_p$ are the parameters of the model, and ε_t is white noise. This can be equivalently written using the backshift operator B as

>> Section 3

$E[(X_{t+n} - \mu)(X_t - \mu)] = \mu[1 - \theta^n] + X_t \theta^n$ and

>> Section 4

An alternative formulation is in terms of the autocorrelation function. The AR parameters are determined by the first $p+1$ elements $\rho(\tau)$ of the autocorrelation function. The full autocorrelation function can then be derived by recursively calculating

>> Section 5

The partial autocorrelation of an AR(p) process equals zero at lags larger than p , so the appropriate maximum lag p is the one after which the partial autocorrelations are all zero.

>> Section 6

where B is the backshift operator, where $\phi(\cdot)$ is the function defining the autoregression, and where ϕ_k are the coefficients in the autoregression. The formula is valid only if all the roots have multiplicity 1. The autocorrelation function of an AR(p) process is a sum of decaying exponentials.

>> Section 7

$X_t = \phi_N X_{t-N} + \sum_{k=0}^{N-1} \phi_k \varepsilon_{t-k}$

>> Section 8

Calculation of the AR parameters There are many ways to estimate the coefficients, such as the ordinary least squares procedure or method of moments (through Yule–Walker equations). The AR(p) model is given by the equation

>> Section 9

$S(f) = \frac{\sigma^2}{|1 - \phi_1 e^{-2\pi i f} - \dots - \phi_p e^{-2\pi i p f}|^2} = \frac{\sigma^2}{|1 - \phi_1 e^{-2\pi i f} - \dots - \phi_p e^{-2\pi i p f}|^2}$

>> Section 10

which can be solved for all ϕ_m ; $m = 1, 2, \dots, p$. The remaining equation for $m = 0$ is

>> Section 11

It is based on parameters ϕ_i where $i = 1, \dots, p$. There is a direct correspondence between these parameters and the covariance function of the process, and this correspondence can be inverted to determine the parameters from the autocorrelation function (which is itself obtained from the covariances). This is done using the Yule–Walker equations.

>> Section 12

When the polynomial division on the right side is carried out, the polynomial in the backshift operator applied to ε_t has an infinite order—that is, an infinite number of lagged values of ε_t appear on the right side of the equation.

Research Notes on Autoregressive model

Autoregressive model

>> Section 13

Impulse response The impulse response of a system is the change in an evolving variable in response to a change in the value of a shock term k periods earlier, as a function of k . Since the AR model is a special case of the vector autoregressive model, the computation of the impulse response in vector autoregression#impulse response applies here.

>> Section 14

$$S(f) = \frac{\sigma^2}{1 - \sum_{k=1}^p \phi_k e^{-i2\pi f k}} \quad \{ \displaystyle S(f) = \frac{\sigma^2}{1 - \sum_{k=1}^p \varphi_k e^{-i2\pi f k}} \}$$

>> Section 15

When $\phi_1 < 0$ $\{ \displaystyle \varphi_1 < 0 \}$ it acts as a high-pass filter on the white noise with a spectral peak at $f = 1/2$ $\{ \displaystyle f = 1/2 \}$. The process is non-stationary when the poles are on or outside the unit circle, or equivalently when the characteristic roots are on or inside the unit circle. The process is stable when the poles are strictly within the unit circle (roots strictly outside the unit circle), or equivalently when the coefficients are in the triangle $-1 \leq \phi_2 \leq 1 - |\phi_1|$ $\{ \displaystyle -1 \leq \varphi_2 \leq 1 - |\varphi_1| \}$. The full PSD function can be expressed in real form as:

>> Section 16

$$\Phi(\omega) = \frac{1}{2\pi} \frac{\sigma^2}{1 - \phi_2 \gamma \pi(\gamma^2 + \omega^2)} \quad \{ \displaystyle \Phi(\omega) = \frac{1}{2\pi} \frac{\sigma^2}{1 - \varphi_2 \gamma \pi(\gamma^2 + \omega^2)} \}$$

>> Section 17

Example: An AR(1) process An AR(1) process is given by: $X_t = \phi X_{t-1} + \varepsilon_t$ $\{ \displaystyle X_t = \varphi X_{t-1} + \varepsilon_t \}$ where ε_t $\{ \displaystyle \varepsilon_t \}$ is a white noise process with zero mean and constant variance σ^2 $\{ \displaystyle \sigma^2 \}$. (Note: The subscript on ϕ_1 $\{ \displaystyle \varphi_1 \}$ has been dropped.) The process is weak-sense stationary if $|\phi| < 1$ $\{ \displaystyle |\varphi| < 1 \}$ since it is obtained as the output of a stable filter whose input is white noise. (If $\phi = 1$ $\{ \displaystyle \varphi = 1 \}$ then the variance of X_t $\{ \displaystyle X_t \}$ depends on time lag t , so that the variance of the series diverges to infinity as t goes to infinity, and is therefore not weak-sense stationary.) Assuming $|\phi| < 1$ $\{ \displaystyle |\varphi| < 1 \}$, the mean $E(X_t)$ $\{ \displaystyle E(X_t) \}$ is identical for all values of t by definition of weak sense stationarity. If the mean is denoted by μ $\{ \displaystyle \mu \}$, it follows from $E(X_t) = \phi E(X_{t-1}) + E(\varepsilon_t)$ $\{ \displaystyle E(X_t) = \varphi E(X_{t-1}) + E(\varepsilon_t) \}$,

$$E(X_t) = \varphi E(X_{t-1}) + E(\varepsilon_t), \text{ that } \mu = \phi \mu + 0, \{ \displaystyle \mu = \varphi \mu + 0, \}$$